To develop a cloud-based AI and data analytics layer that transforms POS-style transactional data into operational insights to support customer segmentation, product performance analysis, waste reduction, and upselling recommendations in hospitality businesses.

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BUSINESS PROBLEM

Hospitality businesses collect large volumes of transactional data through POS systems, but this data is often underused for structured decision-making. Many operational decisions are still based mainly on intuition and experience.

This project explores how POS-style transactional data can be transformed into actionable insights to support customer understanding, product performance analysis, waste reduction support, and upselling opportunities.

PROJECT OBJECTIVES

- Segment customers based on purchasing behaviour.
- Analyse product performance.
- Identify product relationships for upselling.
- Support better stock and waste-related decision-making.
- Demonstrate a cloud-ready deployment using AWS.

METHODOLOGY: CRISP-DM

- Business Understanding — define the problem, objectives, stakeholders, and success criteria.
- Data Understanding — explore the dataset, identify patterns, and assess data quality.
- Data Preparation — clean the data, create Revenue, build RFM features, and prepare the basket matrix.
- Modelling — apply K-Means clustering and Apriori association rules.
- Evaluation — assess whether the models meet business objectives.
- Deployment — export outputs and upload artefacts to AWS S3.

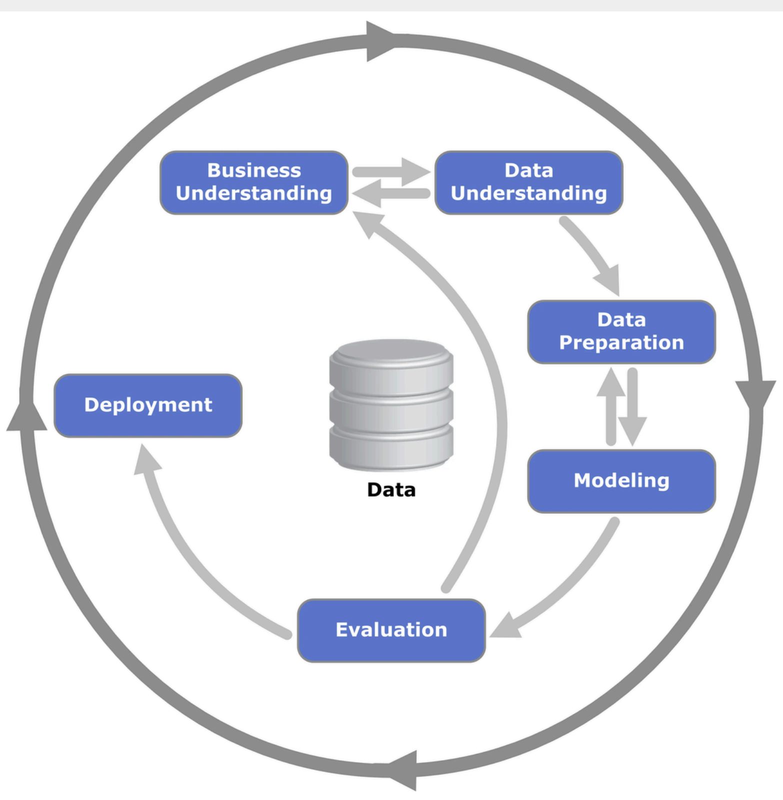
DATASET

Initial dataset:

- 525,461 rows
- 8 columns
- Transactions from December 2009 to December 2010

Cleaned dataset:

- 504,730 rows
- 9 columns
- Revenue feature added



DATA PREPARATION

Main preparation steps:

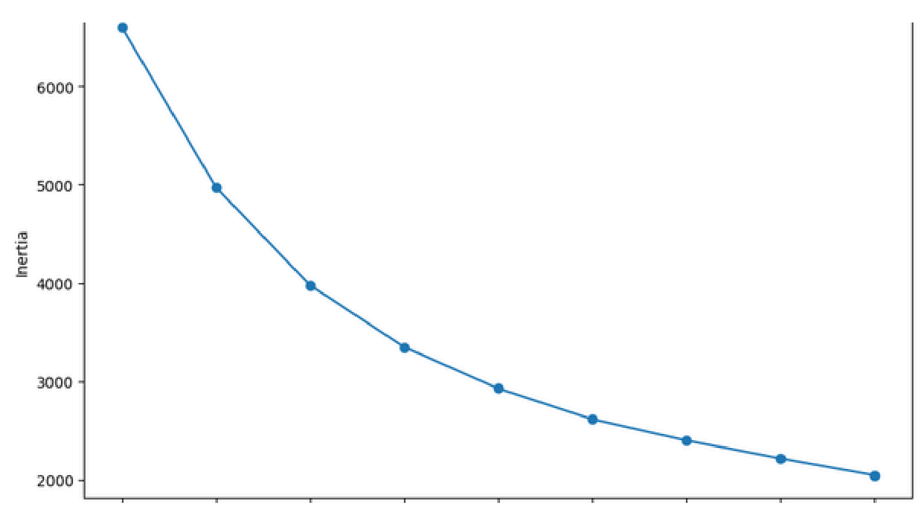
- Removed duplicates and invalid transaction records.
- Removed cancelled invoices, negative quantities, and invalid prices.
- Created Revenue = Quantity × Price.
- Removed non-product records such as postage, manual fees, Amazon fees, bank charges, and gift vouchers.
- Used valid Customer ID records for customer segmentation.
- Used UK-only transactions for market basket analysis.

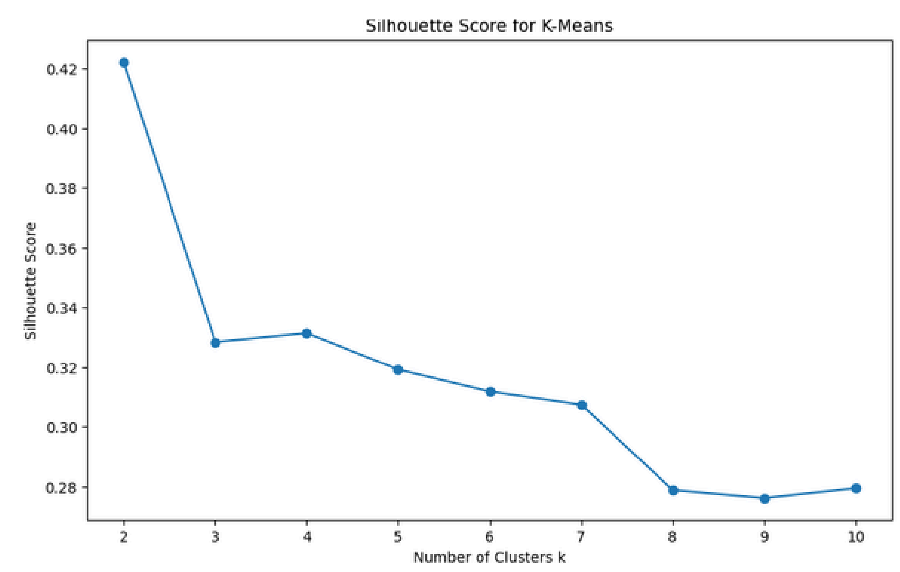
FEATURE ENGINEERING

Two different modelling dataset were made

RFM: Recency, Frequency and Monetary value.

Basket: a Binary basket was created.



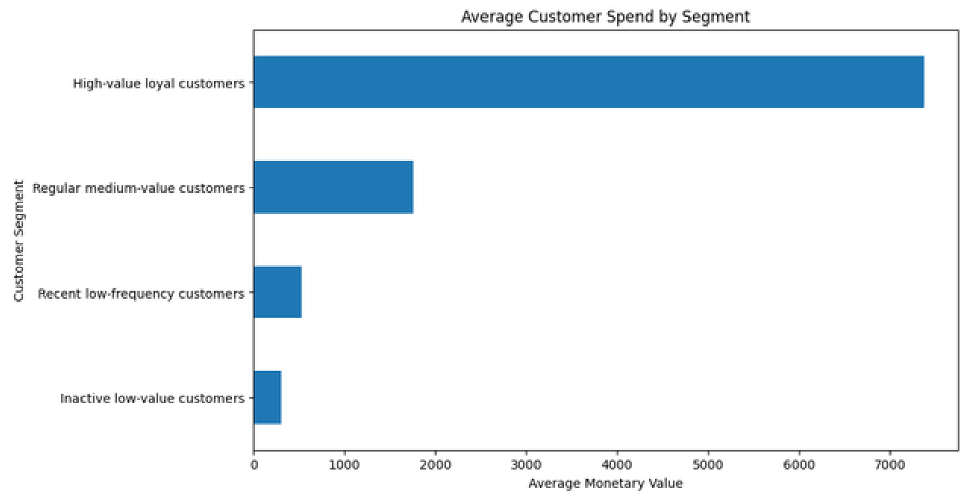


MODELLING A

Two unsupervised techniques were applied.
Model 1: K-Means Clustering
Purpose: Customer segmentation
Input: Scaled RFM dataset
Tested: k values from 2 to 10
Evaluation: Elbow method and silhouette score
Final model: k = 4
The final K-Means model grouped customers into four behavioural segments.

Model 2: Apriori Association Rules
Purpose: Upselling and product recommendation
Input: UK-only binary basket matrix
Minimum support: 0.03
Evaluation metrics: Support, confidence, and lift
The Apriori model identified products frequently purchased together and generated association rules for upselling recommendations.

CUSTOMER SEGMENTATION



These segments support targeted marketing, loyalty campaigns, reactivation strategies, and personalised offers.

PRODUCT PERFORMANCE

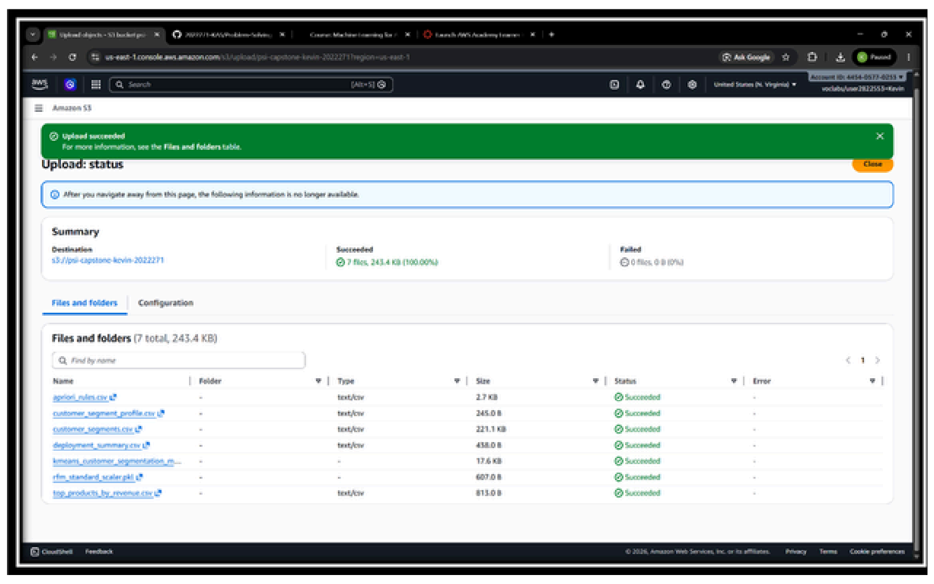
The product performance analysis identified which products generated the highest revenue and which products appeared most frequently in transactions. These insights can support menu optimisation, product prioritisation, and stock planning decisions.

UPSELLING RECOMMENDATION

The association rules can support product bundling, product placement, promotional offers, and cross-selling recommendations.

AWS PROOF-OF-CONCEPT

The AWS S3 upload demonstrates a cloud-ready proof-of-concept where analytical outputs and model artefacts can be stored centrally for future reporting, dashboarding, or application integration.



CONCLUSION

Key outcomes:

- Customer segmentation using K-Means
- Product performance insights
- Upselling recommendations using Apriori
- Cloud-ready storage using AWS S3
- Indirect support for better stock planning and waste reduction

The solution provides a practical proof-of-concept for using data analytics and machine learning to support operational decision-making in hospitality businesses.

KEY REFERENCES

CCT College Dublin CRISP-DM materials; UCI Online Retail II dataset; scikit-learn documentation; mlxtend Apriori documentation; AWS S3 documentation.

Kevin De A. Sanchez - Y4 2022271